



Chapter 1 The main properties of fluids

Section 1 Fluid and its modeling

Section 2 Inertia

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Section 4 Viscosity

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Section 4 Viscosity

- 👉 1, Viscosity and Newton's stress formula
- 2, (Non-)Newtonian fluid
- 3, Flow energy loss





Section 4 Viscosity

1, Viscosity and Newton's stress formula

1, Friction

Friction: The ability of a fluid in **motion** to resist **shear deformation**. It is caused by **molecular attraction** between adjacent fluid layers in liquids and **momentum exchange** between molecules in gases.

Depending on friction, one defines

- a. **Actual fluid:** refers to a fluid that has the ability to resist shear deformation during motion. There is friction.
- b. **Ideal fluid:** refers to an imaginary fluid that is inviscid and cannot resist shearing during motion when deformed. There is no friction.



(1) Definition

Fluid viscosity: Friction is measured by viscosity

(2) Classification

Dynamic viscosity μ : a coefficient reflecting the viscosity of fluid, unit: **N•s/m²**

Kinematic viscosity ν :

$$\nu = \frac{\mu}{\rho} \quad (\text{m}^2/\text{s})$$

The kinematic viscosity of water ν can usually be calculated by the empirical formula:

$$\nu = \frac{0.01775 \times 10^{-4}}{1 + 0.0337T + 0.000221T^2} \quad (\text{m}^2/\text{s})$$





(3) Factors affecting viscosity

μ varies with the type of fluid and with pressure and temperature

- 1) Type of fluid. Generally, the viscosity of a liquid is **greater** than that of a gas.
- 2) Pressure. For common fluids, such as water, gas, etc., the change of μ with pressure is **small** and can generally **be ignored**.
- 3) Temperature. It is the main factor affecting viscosity. As the temperature increases, the viscosity of liquids **decreases** and the viscosity of gases **increases**.



a. Liquid: Molecular attraction is the main factor that produces viscosity. When the temperature increases, the distance between molecules increases and the attractive force decreases, so the shear stress generated by the shear deformation speed decreases, so μ decreases.

b. Gas: The distance between the gas molecules is large and the attractive force is small, so the viscosity is mainly caused by the momentum exchange between the gas molecules. As the temperature increases, the molecular motion accelerates, and the momentum exchange becomes more frequent, so μ increases.





Multiple choice question: Which of the following statements about fluid viscosity is incorrect?

- A. Viscosity is an inherent property of fluids;
- B. Viscosity is a measure of the ability of a fluid in motion to resist the rate of shear deformation;
- C. The viscosity of fluid can both transmit motion and resist motion;
- D. The viscosity of the fluid increases with the increase of temperature.





3, Newton's stress formula (or Newton's law of internal friction)

Newton's stress formula: When the liquid moves, the shear stress generated between adjacent liquid layers is proportional to the rate of shear deformation.

$$\tau = \mu \frac{du}{dy} = \mu \frac{d\theta}{dt} \quad (\text{N/m}^2, \text{Pa})$$

τ —viscous shear stress, which is the internal friction force per unit area.

Explanation: 1) The shear stress of the fluid is proportional to the shear deformation rate or the angular deformation rate - important characteristics that distinguish it from solids. The shear stress of a solid is proportional to the magnitude of the angular deformation.

2) The shear stress of the fluid is proportional to the dynamic viscosity μ

3) For fluid in equilibrium, $du/dy=0$, for ideal fluid, $\mu=0$, no shear stress, $\tau=0$.



Flat-plate experiment

Linear flow velocity profile:

$$u(y) = \frac{U}{Y} y$$

thus

$$\frac{du}{dy} = \frac{U}{Y}$$

Experiments show that for most fluids:

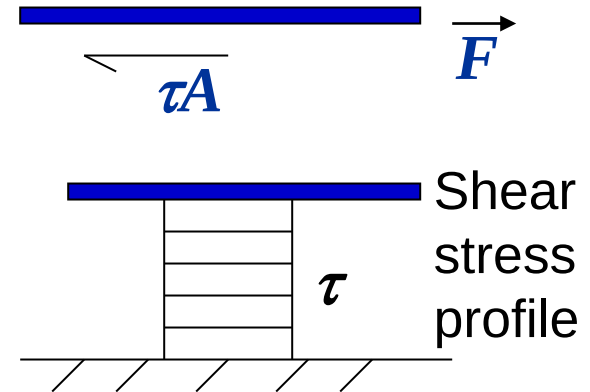
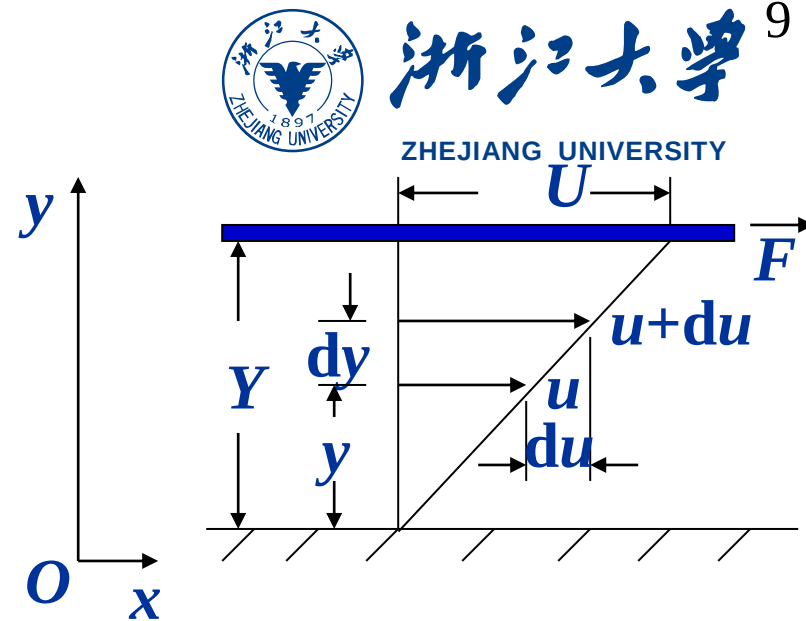
$$F \propto \frac{AU}{Y}$$

Introducing the dynamic viscosity μ , we get **Newton's stress formula**

$$\tau = \frac{F}{A} = \mu \frac{U}{Y} = \mu \frac{du}{dy}$$

where: $\frac{du}{dy}$ represents the shear deformation

rate of the liquid parcel. We can prove: $\frac{du}{dy} = \frac{d\theta}{dt}$





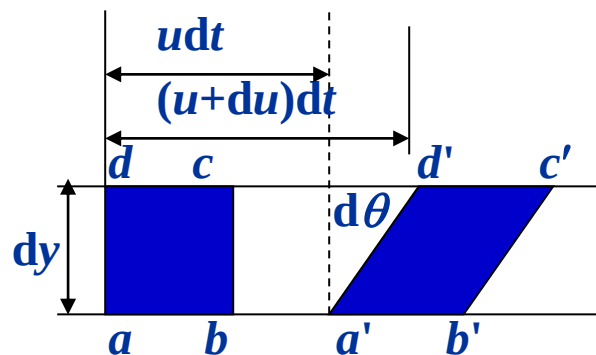
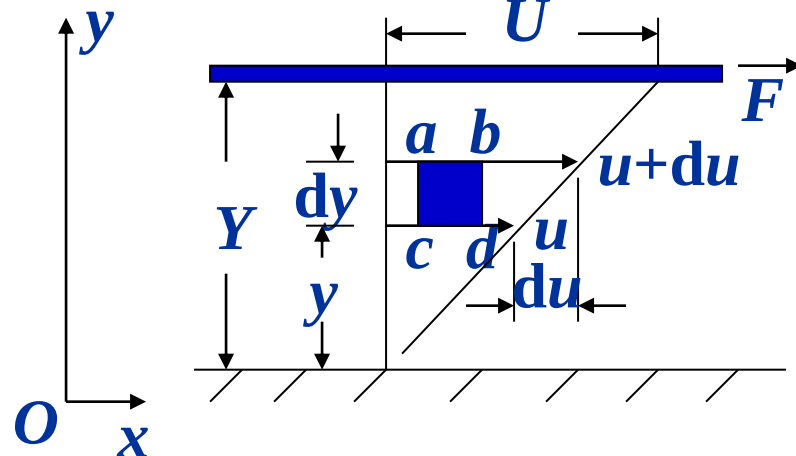
Flat-plate experiment

Proof:

From sketch: $d\theta \approx \tan(d\theta) = \frac{du}{dy}$

$$\frac{d\theta}{dt} = \frac{du}{dy}$$

thus
$$\tau = \mu \frac{du}{dy} = \mu \frac{d\theta}{dt}$$



Description: The shear stress of the fluid is proportional to the shear deformation rate, or the angular deformation rate.



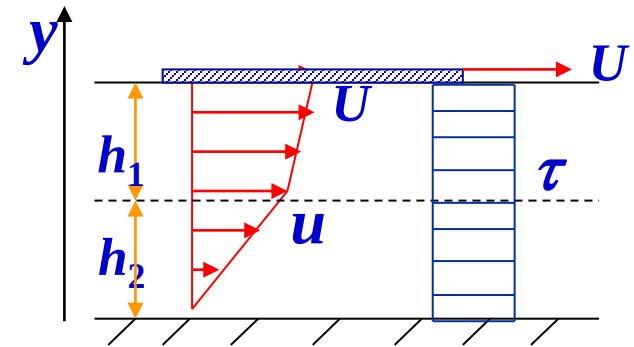
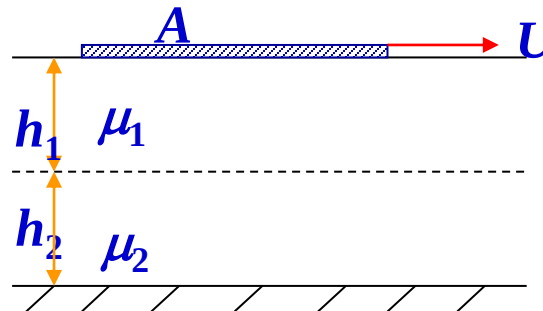
Example 1: Flow velocity and the shear stress profile diagrams of the liquid between the plates. The flow velocity is distributed in a straight line.

Solution: Flow velocity at liquid interface = u , then:

Shear stress profile

Upper layer: $\tau_1 = \mu_1 \frac{U-u}{h_1}$

Lower layer: $\tau_2 = \mu_2 \frac{u-0}{h_2}$



At liquid interface: $\tau = \tau_1 = \tau_2$

$$\therefore u = \frac{\mu_1 h_2 U}{\mu_2 h_1 + \mu_1 h_2}$$

Flow velocity profile:

Upper layer: $u_1 = u + \frac{U-u}{h_1}(y-h_2)$

Lower layer: $u_2 = \frac{u}{h_2} y$

Question: What is larger in figures, μ_1 or μ_2 ?

Answer: Because of $\mu = \frac{\tau}{du/dy}$ ($\tau = C$), $\mu_1 > \mu_2$.

For ideal fluid: $\tau = 0$, $\mu = 0$





Example 2: A wooden block with a bottom area of $0.4 \times 0.45 \text{ m}^2$, a height of 0.01 m , and a mass of 5 kg , moves with constant velocity $u=1 \text{ m/s}$ downward along a slope coated with lubricating oil. The thickness of the oil layer is $\delta = 0.1 \text{ mm}$, the velocity profile of the oil layer driven by the wood block is linear. What is the viscosity μ of the oil?

Constant velocity means:

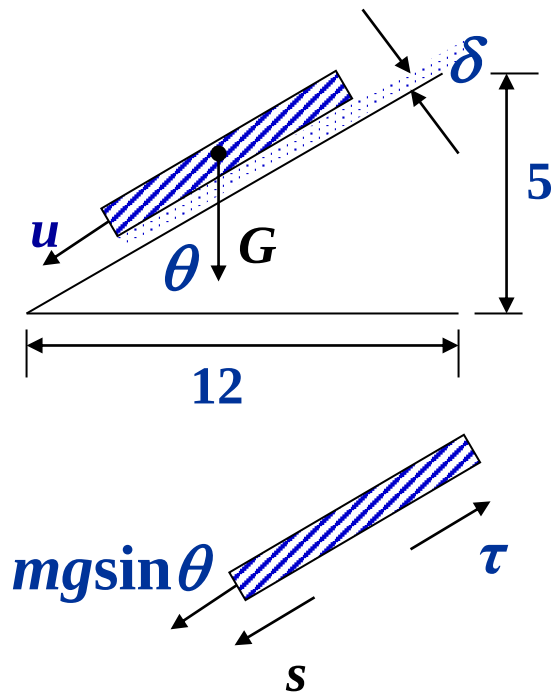
$$\sum F = 0$$

$$mg \sin \theta - \tau \cdot A = 0$$

$$\tau = \mu \frac{du}{dy} = \mu \frac{u}{\delta} \text{ (linear velocity profile)}$$

$$\therefore \theta = \tan^{-1}(5/12) = 22.62^\circ$$

$$\therefore \mu = \frac{mg \sin \theta \cdot \delta}{Au} = 0.0105 \text{ N} \cdot \text{s} / \text{m}^2$$



Example 3: A disc with a diameter of **0.1 m** is driven by a shaft rotating on a platform. They are separated by an oil film with a thickness of **$\delta=1.5$ mm**. When the disc rotates at **$n=50/\text{min}$** , the measured torque is **$M = 2.94 \times 10^{-4} \text{ N}\cdot\text{m}$** . Assuming a linear velocity profile along the vertical direction, find **μ** and **τ_0** .

Solution: $u = \omega r = 2\pi n r$

$$dT(\text{dF in book}) = \mu dA \frac{u}{\delta} = \mu \cdot 2\pi r dr \cdot \frac{1}{\delta} \cdot 2\pi n r$$

The viscous frictional resistance torque of the disc element is: **$dM = dTr = 4\mu\pi^2 r^3 n dr / \delta$**

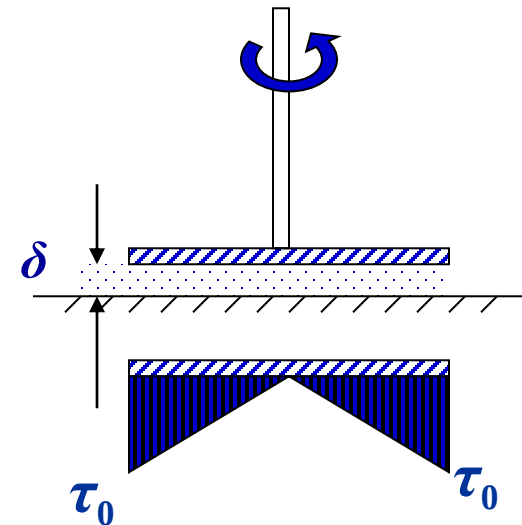
Then the total friction torque to overcome is:

$$M = \int_0^R dM = 4\mu\pi^2 \frac{n}{\delta} \int_0^R r^3 dr = \frac{\mu\pi^2 n R^4}{\delta}$$

$$\mu = \delta M / (\pi^2 n R^4) = 85.8 \times 10^{-4} \text{ Pa}\cdot\text{s}$$

$$\tau_0 = \mu \frac{du}{dy} = \mu \frac{\omega R}{\delta} = \mu \cdot \frac{2\pi n}{\delta} R \quad (R = 0.05 \text{ m})$$

$$\therefore \tau_0 = 1.50 \text{ Pa}$$





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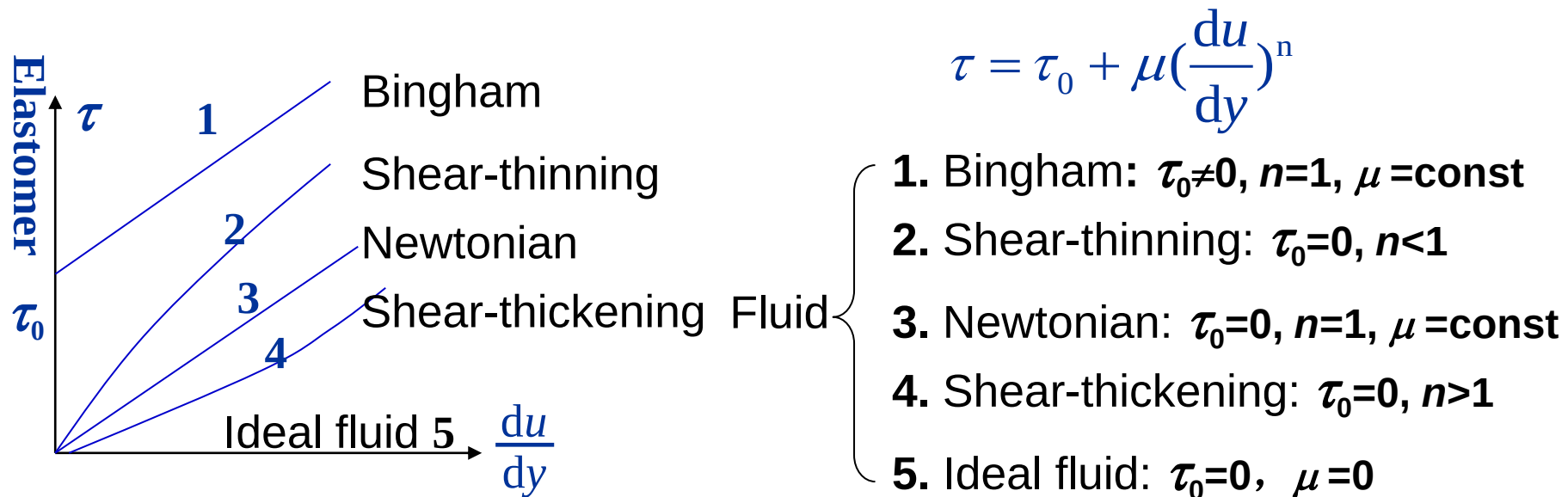
- 1, Viscosity and Newton's stress formula
- ☞ 2, (Non-)Newtonian fluid
- 3, Flow energy loss



2, (Non-)Newtonian fluid

Newtonian Fluids: Fluid that obeys Newton's stress formula

Non-Newtonian Fluids: Fluid that does not obey Newton's stress formula



Question: The fluid in which the shear stress is linearly related to the shear deformation rate is a Newtonian fluid, right?



Fluid classification



Fluid category			Definition	$\tau = \tau_0 + \mu(\frac{du}{dy})^n$	Examples
Ideal fluid			Hypothetical fluid with no viscosity	$\mu = 0$ 、 $\tau_0 = 0$	
Actual fluid	Newtonian fluid		Viscous, compressible fluid, $\mu \neq 0$	Obeys Newton's stress formula, $\tau_0 = 0$ 、 $\mu \neq 0$ 、 $n = 1$	Water, air, gasoline, kerosene, toluene, ethanol, etc
	Non-Newtonian fluid	Bingham fluid		$\tau_0 \neq 0$ 、 $\mu = \text{const}$ 、 $n = 1$	Toothpaste, mud, blood, etc
		Shear-thinning fluid		$\tau_0 = 0$ 、 $\mu \neq 0$ 、 $n < 1$	Rubber, paint, nylon, etc
		Shear-thickening fluid		$\tau_0 = 0$ 、 $\mu \neq 0$ 、 $n > 1$	Dough, thick starch paste
Compressible fluid			$\rho = \rho(x, y, z, t)$		
Incompressible fluid			$\rho = \text{const}$		





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3, Flow energy loss

In the process of fluid flow, the **internal friction force** works and continuously consumes the mechanical energy of the fluid flow, which is converted into **heat energy and dissipated**. The **mechanical energy consumption** of this fluid flow is called fluid flow energy loss.





Example: dumping water from a height of **$h=100\text{m}$** to the ground. Imagine the energy is completely converted into heat energy and all the heat energy is absorbed by the water body. How much will the water body heat up (specific heat capacity of water **$C = 4184 \text{ J/(kg K)}$**) ?

Solution: Potential energy $\rightarrow$ heat

M is mass of water:

$$E_{\text{pot}} = Mgh$$

$$Q = CM \Delta T$$

$$Q = E_{\text{pot}}$$

$$\Delta T = gh/C = 0.2342 \text{ K}$$





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- 1, Cohesion, adhesion, surface tension
- 2, Capillary action
- 3, Calculation of capillary action





Section 5 Surface tension

1, Cohesion, adhesion, surface tension

Cohesive Force: Attraction between molecules of a substance

Adhesion: Attraction between molecules of two different substances in contact

Surface Tension: the energy or work required to increase the surface area of a liquid due to intermolecular forces, or the tendency of liquid surfaces at rest to shrink into the minimum surface area possible. Mercury and water are the liquids with the highest surface tensions.

Surface tension coefficient σ [N/m]: the surface tension per unit length on the free liquid surface.





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1, Cohesion, adhesion, surface tension

☞ 2, Capillary action

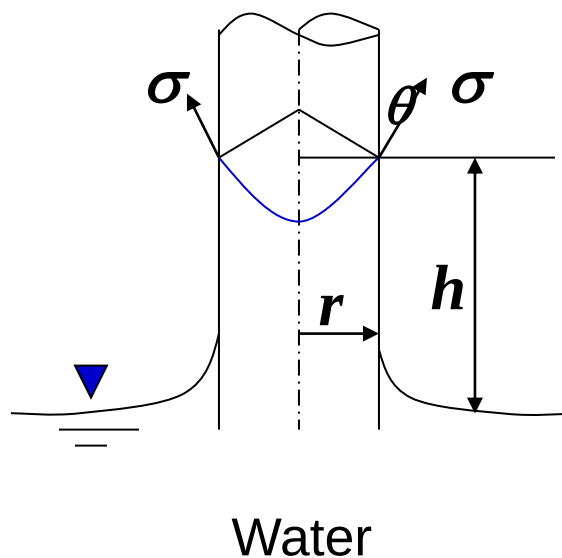
3, Calculation of capillary action



2, Capillary action



Capillary action: the phenomenon that, when an object containing fine gaps is in contact with liquid, **the liquid rises** or penetrates along the gap if wetted, and the liquid falls along the gap if not wetted.





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- 👉 3, Calculation of capillary action



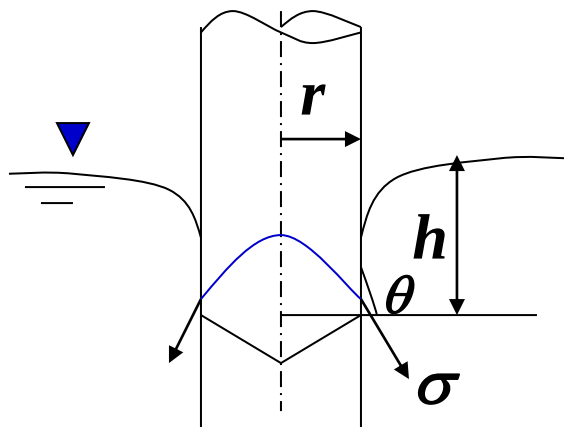
3, Calculation of capillary action

Capillary height:

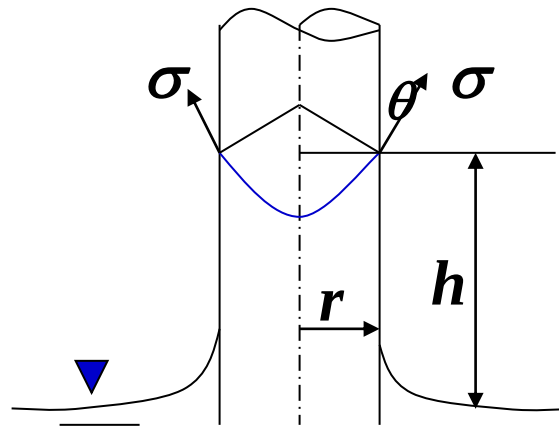
$$h = \frac{2\sigma \cos \theta}{\rho g r}$$

For water: $\theta = 0^\circ$, $\sigma = 0.074 \text{ N/m}$

For mercury: $\theta = 140^\circ$, $\sigma = 0.514 \text{ N/m}$



Mercury



Water





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Section 6 Vapor pressure



1, Vaporization and condensation

Vaporization: the process in which molecules escape from the liquid and diffuse into the space as gas

The inverse process is called **Condensation**. It increases with vapor pressure.

2, Saturated vapor pressure (Vaporization pressure)

Vaporization pressure: the vapor pressure needed for condensation to compensate vaporization. It increases with temperature. In a closed system, it can be measured as the saturated vapor pressure at which vaporization and condensation are in dynamic equilibrium. It is a measure for the tendency of a liquid to vaporize.

Question: What happens at the boiling point of a liquid in an open system?

Answer: The vaporization pressure overcomes the atmospheric pressure.





3, Cavitation

Cavitation: a phenomenon in which the **liquid pressure** reduces to below the **liquid's vapor pressure**, leading to the formation of small vapor-filled cavities (“bubbles” or “voids”) in the liquid.

4, Cavitation erosion

Cavitation erosion: When subjected to **higher** pressure, the cavities **collapse**, which can generate **shock waves** that may damage machinery. These shock waves are **strong** when they are **very close** to the imploded bubble, but rapidly **weaken** as they propagate away from the implosion.





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Section 7 Forces acting on fluid

1, Classification

- 1, based on physical property: **gravity, friction, elastic force, surface tension**
- 2, based on mode of action: **mass force, surface force**

2, Mass Force:

- 1, **Mass Force**: the force acting on **each fluid particle**. It is proportional to the fluid mass or, in case of a homogeneous fluid, to the fluid volume (and is therefore also called **volume force**).
- 2, **Force per unit mass**: the mass force experienced by a unit mass of fluid

$$\vec{f} = \frac{\vec{F}}{M} = f_x \vec{i} + f_y \vec{j} + f_z \vec{k} = \frac{F_x}{M} \vec{i} + \frac{F_y}{M} \vec{j} + \frac{F_z}{M} \vec{k}$$

Unit: **[m/s²]** = **unit of acceleration**

The most common mass forces are **gravity** and **inertial forces** (e.g., the Coriolis force).





Multiple-choice question: Compare the magnitude of the force per unit mass f_{water} and f_{mercury} on water and mercury in the gravitational field (mass force is only gravity)?

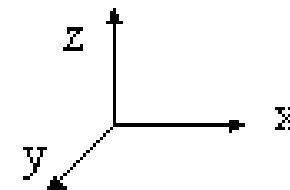
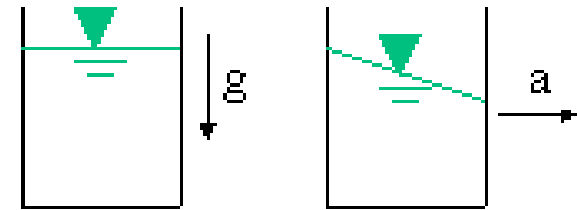
A, $f_{\text{water}} < f_{\text{mercury}}$; **B**, $f_{\text{water}} > f_{\text{mercury}}$; **C**, $f_{\text{water}} = f_{\text{mercury}}$; **D**, unknown

Thinking question: How much is the force per unit mass (f_x, f_y, f_z) of a liquid in a container in the state of free fall and under acceleration a moving in the x -direction?

Static container: $f_x = 0, f_y = 0, f_z = -g$;

Free fall: $f_x = f_y = f_z = 0$;

Acceleration: $f_x = -a, f_y = 0, f_z = -g$



坐标系



3, Surface force

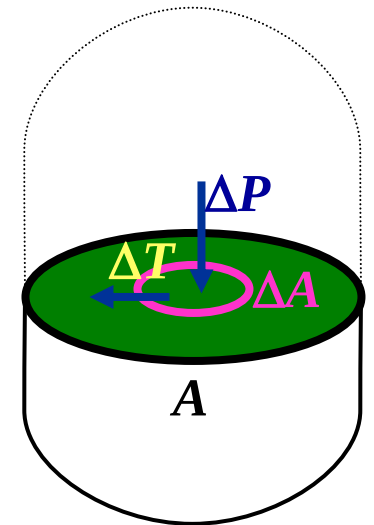
1, **Surface Force**: a directly exerted contact force acting on the surface of an object by an adjacent fluid or other object. Its size is proportional to the area of the surface of action.

Two types of surface forces:

- { Pressure: perpendicular to the surface of action
- { Shear force: parallel to surface of action

2, **Stress**: Surface force per unit area, unit: **N/m²** or **Pa**

$$\left\{ \begin{array}{l} \text{Pressure} \quad p = \lim_{\Delta A \rightarrow 0} \frac{\Delta P}{\Delta A} \\ \text{Shear stress} \quad \tau = \lim_{\Delta A \rightarrow 0} \frac{\Delta T}{\Delta A} \end{array} \right.$$





1, Does the ideal fluid have energy loss? Why?

No. Because of the ideal fluid, $\mu=0$, there is no shear stress.

2, The shear stress of fluids is related to ___, that of solids is related to ___ .

Shear deformation rate; shear deformation magnitude

3, What determines viscosity? How does it change with T ?

Type of fluid, temperature, pressure.

Liquids: decrease with T ; gases: increase with T .





4, τ of a Newtonian fluid increases linearly with **du/dy** , so when τ increases linearly with **du/dy** , the fluid must be a Newtonian fluid?

Not necessarily, because μ increases linearly with du/dy for Bingham plastic fluids, but the curve does not pass through the origin.

5, Why are the dewdrops on lotus leaves always spherical?
Because of their special microstructure and the surface tension of water.

6, A towel with one end placed in the water in the washbasin and one end outside the washbasin, after a period of time, the table outside the washbasin got wet, why?

Because of capillary action.





7, Why is the diameter of a pressure-measuring tube usually not less than **1cm**?

If the inner passage of the tube is small, it will cause capillary action, and the height of the liquid level rising or falling in the capillary tube is relatively large, which will cause excessive errors.

8, If the reading of the pressure tube is h_1 and the capillary height is h_2 , what is the actual height of the pressure measuring tube at this point? (for water and mercury)

$h_1 - h_2$ —— water

$h_1 + h_2$ —— mercury





9, Why do you need to cover the pot when you boil eggs on a plateau **in China**?

On a plateau, the atmospheric pressure is low and the water will boil well below **100° C**. **In China**, this is apparently considered a problem, so the pot is covered. **In Europe**, we just cook longer.

10, Describe the reasons for cavitation of turbine blades

Bubbles are generated at low pressure, and the bubbles follow the water flow to high pressure regions, resulting in an impact force, eroding the blades, and forming cavitation.





11, Can a liquid reach a complete vacuum state (zero pressure)? If not, what is the maximum vacuum?

No, the maximum vacuum is equal to the difference between atmospheric pressure and vaporization pressure.

12, What kinds of forces act on a stationary fluid?

Gravity, compressive stress, but no inertial and shear forces.

13, Which forces act on a fluid in motion?

Gravity, inertial force, compressive stress, shear force.

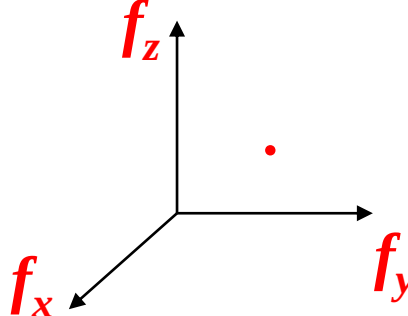




14, Which forces act on an ideal fluid in motion?

Gravity, inertial force, compressive stress, no shear force.

15, What is the force per unit mass of a static fluid with only gravity acting on it? (The coordinate axis z is in the vertical direction and directed upward).

$$f_x = f_y = 0, \quad f_z = -g$$






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1, **Fluid Mechanics** is the study of the **macroscopic mechanical motion** of fluids. When fluids are at rest, they cannot resist shear deformation and will deform or flow under any **slight shear stress**. If the fluid is regarded as a continuous medium without voids, all physical quantities in the fluid (such as velocity $\mathbf{u}$ and density ρ) can be regarded as continuous functions of space and time.

2, The compressibility of a fluid can generally be described by the volume compressibility β and the volume elastic modulus K . Usually, when pressure variations are small, a fluid can be regarded as incompressible.

$$\kappa = -\frac{dV/V}{dp} = \frac{d\rho/\rho}{dp}$$
$$K = \frac{1}{\kappa} = -\frac{dp}{dV/V} = \frac{dp}{d\rho/\rho}$$





3, The **internal friction** of a fluid describes how well it **resists shear deformation**. It is quantified by the dynamic viscosity coefficient μ or kinematic viscosity coefficient ν . It is **affected by temperature**: as the temperature increases, the viscosity of a gas **increases** and the viscosity of a liquid **decreases**.

4, Newton's stress formula (or Newton's law of internal friction)
$$\tau = \mu \frac{du}{dy} = \mu \frac{u}{\delta}$$

It means that the magnitude of the **fluid shear stress is proportional to the velocity gradient** or the rate of angular deformation or the rate of shear deformation, which is an important characteristic of the **fluid that distinguishes it from the solid** (whose shear stress is proportional to the magnitude of the shear deformation). Fluids can be divided into Newtonian fluids and non-Newtonian fluids according to whether they obey Newton's stress formula.





5, Due to capillary cation caused by surface tension, the diameter of the pipe used as a pressure measuring tube is not less than 10mm.

$$h = \frac{29.8}{d} \text{ mm} \quad \text{——Water height rises in pipe}$$

$$h = \frac{10.15}{d} \text{ mm} \quad \text{——Mercury height drops in tube}$$

6, Vaporization pressure is the pressure at which the liquid and gas phases reach dynamic equilibrium. When the external pressure (e.g., from the atmosphere) is equal to or slightly lower than the vaporization pressure, the liquid boils.

7, Forces acting on the fluid: mass force and surface force; the most common mass forces are gravity and the inertial force, and surface force is often divided into pressure perpendicular to the surface and shear force parallel to the surface.

